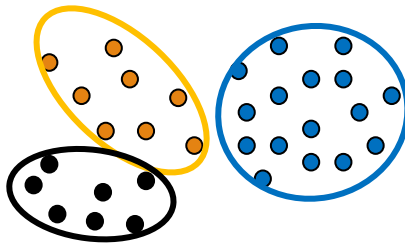


Determining the Number of Clusters

- ❑ The appropriate number of clusters controls the proper granularity of cluster analysis
 - ❑ Finding a good balance between compressibility and accuracy in cluster analysis
- ❑ Two undesirable extremes
 - ❑ The whole data set is one cluster: No value of clustering
 - ❑ Treating each point as a cluster: No data summarization

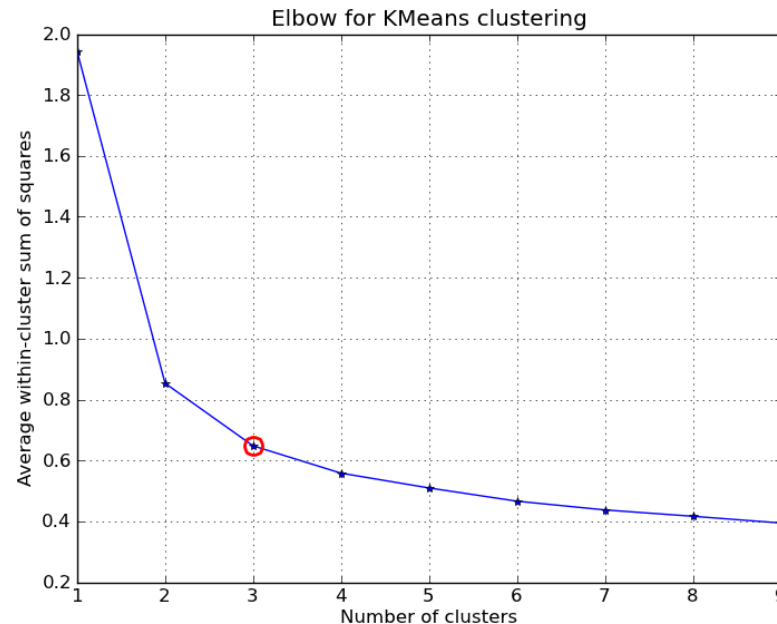


Determining the Number of Clusters

- ❑ The right number of clusters often depends on the distribution's shape and scale in the data set, as well as the clustering resolution required by the user
- ❑ Methods for determining the number of clusters
- ❑ An empirical method
 - ❑ # of clusters: $k \approx \sqrt{\frac{n}{2}}$ for a dataset of n points (e.g., $n = 200$, $k = 10$)
 - ❑ Each cluster is expected to have about $\sqrt{2n}$ points

Finding the Number of Clusters: the Elbow Method

- ❑ Use the turning point in the curve of the sum of within cluster variance with respect to the # of clusters
 - ❑ Increasing the # of clusters can help reduce the sum of within-cluster variance of each cluster
 - ❑ But splitting a cohesive cluster gives only a small reduction



Finding K, the Number of Clusters: A Cross Validation Method

- ❑ Divide a given data set into m parts, and use $m - 1$ parts to obtain a clustering model
- ❑ Use the remaining part to test the quality of the clustering
 - ❑ For example, for each point in the test set, find the closest centroid, and use the sum of squared distance between all points in the test set and their closest centroids to measure how well the model fits the test set
- ❑ For any $k > 0$, repeat it m times, compare the overall quality measure w.r.t. different k 's, and find # of clusters that fits the data the best